

MODEL QUESTION PAPER

TED(10) 4085

(Revision 2010)

FIFTH SEMESTER DIPLOMA EXAMINATION
(COMMON TO EI &IT)

INDUSTRIAL INSTRUMENTS II

Time : 3 hrs

(Maximum marks: 100)

PART - A

(Answer all questions Each carry 2 marks)

I

1. Write the applications of Bernoulli's Theorem
2. Draw the sketch of a Flow nozzle
3. Mention the Advantages of Electromagnetic flow meter
4. Define Dew point
5. List the classifications of Tachometer

(5 x 2 =10)

PART - B

(Maximum marks = 30)

II. Answer any 5 questions – Each question carries 6 marks

1. Compare Laminar & Turbulent flow
2. Calculate the volume flow rate in LPM . Water is flowing through pipe & venture meter is used to measure the volumetric flow
diameter(pipe)=0.1m ,diameter(throat)=0.05m
difference in Hg level in the Hg manometer=150mm of Hg
Cd=0.98
3. Water is flowing through the rectangular notch .Height of water from the sill is given by 120mm . breadth of notch is 15cm Cd is given by .68 Calculate the discharge through notch in LPH
4. Describe the working & Construction of ultrasonic flowmeter
5. Check whether the most suitable method of specific Gravity measurement is weighing tube method or Hydrometer . Justify
6. Write the operation of Dew cell
7. Describe any method of Acceleration Measurement

(5×6 = 30)

PART - C

(Answer one full question from each module. Each carries 15 marks)

Module I

- III a) Explain the construction & working of Rotameter (8)
b) Illustrate the DP type flowmetering (7)

OR

- IV. a) Distinguish between variable area flow meter & variable head flow meter (8)
b) Explain about different types of Tappings (7)

Module II

- V. a) Explain the working of any positive Displacement Flowmeter (8)
b) Explain the measurement of flow using hot wire Anemometer (7)

OR

- VI. a) Mention the Advantages and disadvantage of a turbine flowmeter (5)
b) Compare weirs and Notches (6)
c) State coriolis principle (4)

Module III

- VII. a) Using the schematic diagram explain the measurement of dewpoint by cold mirror method (9)
b) Define the following term & mention their units (2x3=6)
1) Absolute humidity
2) Specific gravity
3) Relative humidity

OR

- VIII. a) Write short notes on 1) Saybolt viscometer
2) Redwood viscometer (4X2=8)
b) Using the schematic diagram explain the operation of Wet and dry bulb psychrometer (7)

Module IV

- IX. a) With a neat sketch Describe construction and working of mechanical tachometer (8)
b) Write the advantage of LVDT accelerometer (3)
c) Write short notes on Stroboscope (4)

OR

- X. a) Explain the working of Absorption type Dynamometer (6)
b) Describe the working of drag cup Tachometer (6)
c) Mention the features of piezoelectric accelerometer (3)